

ENVX2001 Lab 07 - Regression modelling

ENVX2001 Applied Statistical Methods

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Welcome to week 7!

This week's lab will give you some practice fitting and interpreting Simple Linear Regression (SLR) and Multiple Linear Regression (MLR) models.

This will prepare you for next week's content where we will seek to improve our models, and then in week 9 we will use improved models for prediction.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use of the statistical programming language R to apply an ANOVA and fit regression models to experimental data
- LO5. interpret the output and understand conceptually how it's derived of a regression, ANOVA and multivariate analysis that have been calculated by R
- LO6. write statistical and modelling results as part of a scientific report

Specific goals

By the end of this lab, students should be able to:

- ☐ Fit a simple linear regression model
- ☐ Assess model assumptions and apply transformations when needed
- ☐ Interpret model parameters and fit statistics
- ☐ Fit a multiple linear regression model (**different structure**)
- ☐ Assess model assumptions, recalling key differences between MLR and SLR (**collinearity**)
- ☐ Determine the most suitable fit statistics for assessing the fit of a MLR model (**r-sq vs adj**)
- ☐ Write a conclusion on your findings



Tip

Please work on this exercise by creating your own R Markdown file.

Preparation

- ☐ Install or update the performance package

CODE

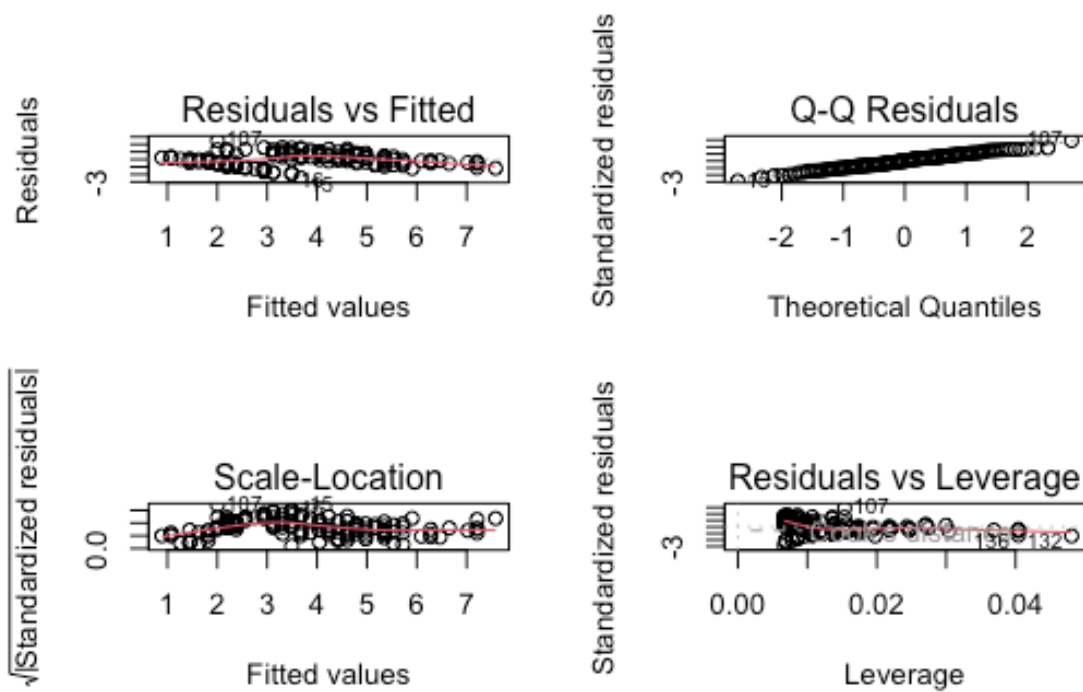
```
#install.packages("performance")
library(performance)
```

This package is really good for checking your models. For this lab, we will focus on the `check_model()` function, which gives us nice pretty diagnostic plots for models:

`plot()`

CODE

```
par(mfrow=c(2,2))
plot(iris_lm)
```



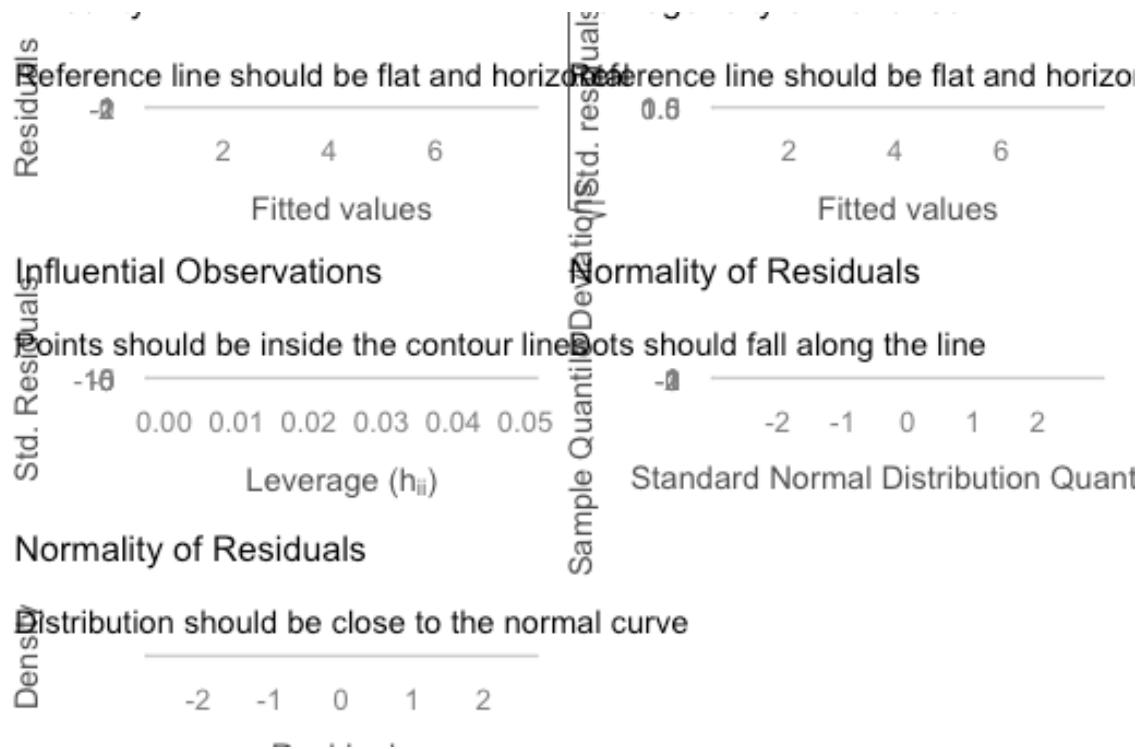
CODE

```
par(mfrow=c(1,1))
```

check_model() from performance

CODE

```
library(performance)
check_model(iris_lm)
```



Downloads

File	Used in	Download
Loyn_SLR.xlsx	Exercise 1	Download
california_streamflow.xlsx	Exercise 2	Download

Simple linear regression model fitting and interpretation (~ 30 minutes)

Habitat fragmentation resulting from agriculture and urbanisation has increased the rate of wildlife population decline in Australia. You are a consultant and have been asked to investigate the influence of forest fragment size upon bird abundance. You have been provided with some data containing data on bird abundance and patch area from 56 different forest patches in Southeastern Victoria, Australia, and you want to know if there is a relationship between these two variables.

The data is in the **Loyn_SLR.xlsx** file.

Variables are as follows:

- ABUND: Bird abundance within each forest patch (number of birds)
- AREA: Area of each forest patch (ha)

Original data source: Loyn, R. H. (1987). Effects of patch area and habitat on bird abundances, species numbers, and tree health in fragmented Victorian forests. In *Nature Conservation: the role of remnants of native vegetation* eds. D.S. Saunders, G.W. Arnold, A.A. Burbridge, & A.J.M. Hopkins, pp. 65-77. Surrey Beatty and Sons.



Figure 1: Kookaburra sitting on a branch. Photo by [pen_ash on Unsplash](#)

Analysis

Exercise 1

- Write out the statistical model for a simple linear regression and define it in terms of influence of forest fragment size upon bird abundance. You can write it out in words and/or in mathematical notation.
- State the hypotheses you will be testing for the relationship between bird abundance and patch area.
- Conduct some exploratory data analysis upon your two variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

(iv) Investigate the relationship between bird abundance and patch area. Describe the relationship, supported by values (correlation coefficient) and plots.

(v) Fit a simple linear regression model to the data, with bird abundance as the response variable and patch area as the predictor variable.

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint 1: you can use the `check_model()` function from the `performance` package to check the assumptions of your model.

Hint 2: If you do choose to transform a variable you will need to refit the model and check the assumptions again.

(vii) Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between bird abundance and patch area?

(viii) Write a concluding statement on your findings. Hint: Consider the initial aim and hypothesis of the study and write it in a way that is easily understood by your client.

Multiple linear regression (~ 30 minutes)

You are a water scientist tasked with analysing legacy stream runoff data to determine-precipitation sites influence the stream runoff volume.

The `california_streamflow` dataset contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. I have reduced this to three variables:

- `lake_sabrina`: Precipitation recorded at Lake Sabrina (mm)
- `pine_creek`: Precipitation recorded at Big Pine Creek (mm),
- `rock_creek`: Precipitation recorded at Rock Creek (mm)
- `runoff_volume`: Stream runoff volume measured at a site near Bishop, California (ML/year).

This will be the response variable.

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

This week we will focus on getting the model running and practice interpretation of the output. Next week we will focus on working out the most suitable model.

DATA SOURCE

Analysis

Exercise 1

(i) Write out the statistical model for a simple linear regression and define it in terms of influence of forest fragment size upon bird abundance. You can write it out in words and/or in mathematical notation.

(ii) State the hypotheses you will be testing for the relationship between bird abundance and patch area.

(iii) Conduct some exploratory data analysis upon your two variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

(iv) Investigate the relationship between bird abundance and patch area. Describe the relationship, supported by values (correlation coefficient) and plots.

(v) Fit a simple linear regression model to the data, with bird abundance as the response variable and patch area as the predictor variable.

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint 1: you can use the `check_model()` function from the `performance` package to check the assumptions of your model.

Hint 2: If you do choose to transform a variable you will need to refit the model and check the assumptions again.

(vii) Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between bird abundance and patch area?

(viii) Write a concluding statement on your findings. Hint: Consider the initial aim and hypothesis of the study and write it in a way that is easily understood by your client.

CODE

```
# read in the data
stream_data <- read_xlsx("data/california_streamflow.xlsx", "streamflow")

names(stream_data)
```

OUTPUT

```
[1] "lake_sabrina" "pine_creek"   "rock_creek"   "runoff_volume"
```

CODE

```
s.mod_full <- lm(runoff_volume ~ lake_sabrina + pine_creek + rock_creek, data=stream_data)
s.mod_full <- lm(runoff_volume ~ ., data=stream_data) ## you can also use the . to indicate use all
variables
summary(s.mod_full)
```

OUTPUT

```
Call:
lm(formula = runoff_volume ~ ., data = stream_data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.09885 -0.03331  0.01025  0.03359  0.09495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.25716    0.12360  26.352 < 2e-16 ***
lake_sabrina  0.05631    0.03756   1.499  0.14185
pine_creek   0.21085    0.06756   3.121  0.00339 **
rock_creek   0.43838    0.08798   4.983 1.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04861 on 39 degrees of freedom
Multiple R-squared:  0.8817,    Adjusted R-squared:  0.8726
F-statistic: 96.88 on 3 and 39 DF,  p-value: < 2.2e-16
```

Partial F-Tests

The above analysis tells us that both pine_creek & rock_creek are significant, according to the t-test, in the model and lake_sabrina is not? This involves performing Partial F-Tests as discussed in the lecture.

This can be done in **R** by using anova() on two model objects. To be able to compare the models and run the anova, you need to make objects of all the possible model combinations you want to compare.

```
CODE
s.mod_reduced <- lm(runoff_volume ~ rock_creek + pine_creek, data=stream_data)
anova(s.mod_reduced, s.mod_full)
```

The last row gives the results of the partial F-test.

Question 1

Should we remove lake_sabrina from the model?

Question 2

Is the p-value for the f-test the same as for the t-test?

Question 3

Write out the hypotheses you are testing.

Perform a Partial F-Test to work out if the removal of lake_sabrina and pine_creek improves upon the full model.

```
CODE
s.mod_reduced2 <- lm(runoff_volume ~ lake_sabrina + pine_creek, data=stream_data)
anova(s.mod_reduced2, s.mod_full)
```


OUTPUT

Analysis of Variance Table

```
Model 1: runoff_volume ~ lake_sabrina + pine_creek
Model 2: runoff_volume ~ lake_sabrina + pine_creek + rock_creek
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	40	0.150845				
2	39	0.092166	1	0.05868	24.83	1.321e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Question 4

Which variable should be added to the model containing rock_creek?

Question 5

Could things be even simpler? Perform a partial F-Test to see if a model containing rock_creek alone could be suitable.

CODE

```
s.mod_reduced3 <- lm(runoff_volume ~ rock_creek, data=stream_data)
anova(s.mod_reduced3, s.mod_full)
```

Question 6

What is your optimal model?

Review

- Simple linear regressions model the relationship between two variables
 - We can also make linear models with more than one predictor
- We can use histograms and correlation matrices to do some preliminary exploration of the data
- If any variables are skewed, we can transform them
- Looking at a correlation matrix to identify the best predictors (for both simple and multiple linear regression)
- Fit model using `lm()` function
- Check assumptions:
 - Collinearity (multiple linear regression only)
 - Linearity
 - Independence
 - Normality
 - Equal variance
- Use `summary()` to look at model output and interpret it
 - F-test : overall model significance

- Coefficients table : individual predictors' significance
- R^2 : How much variation in the data is explained by the model?

That's it for today! Great work fitting simple and multiple linear regression! Next week we jump into stepwise selection!

Attribution